OBJECTIVE 3–1 GENERAL CIRCUIT ANALYSIS (SECT. 3–1 TO 3–2)

Given a circuit consisting of linear resistors and independent sources:

(a) (Formulation) Write node-voltage or mesh-current equations for the circuit.

(b) (Solution) Solve the equations from part (a) for selected signal variables or input-output relationships.

Node-voltage method:
See Examples 3–1, 3–2, 3–3, 3–4, 3–5, 3–6 and Exercises 3–2, 3–3, 3–4, 3–5, 3–6

Mesh-current method:
See Examples 3–7, 3–8, 3–9 and Exercises 3–8, 3–9, 3–10

3–1 **(a)** Formulate node-voltage equations for the circuit in Figure P3–1.
(b) Use these equations to find v_x and i_x.

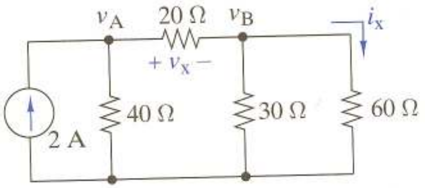

FIGURE P3–1

3–3 **(a)** Formulate node-voltage equations for the circuit in Figure P3–3.

(b) Use these equations to find v_x and i_x.

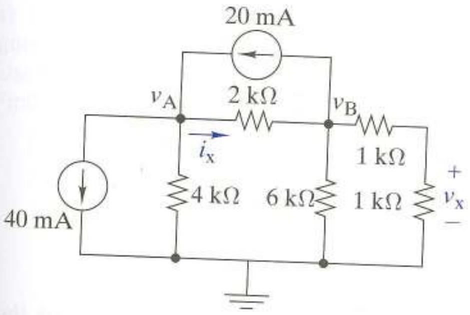

FIGURE P3–3

3-4 **(a)** Formulate node-voltage equations for the circuit in Figure P3–4.

(b) Use these equations to find v_x and i_x.

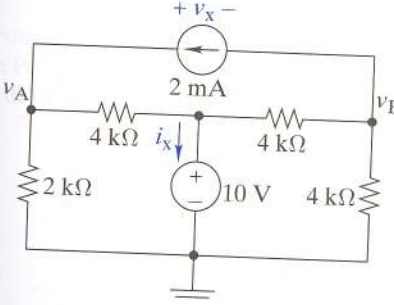

FIGURE P3–4

3–8 **(a)** Formulate node-voltage equations for the circuit in Figure P3–8.

(b) Solve for v_x and i_x when $R_1 = R_4 = 1\ \text{k}\Omega$, $R_2 = R_3 = 250\ \Omega$, $R_x = 500\ \Omega$, and $v_S = 15$ V.

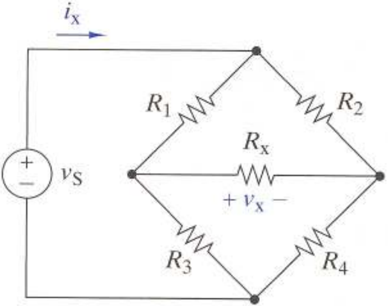

FIGURE P3–8

3–10 **(a)** Formulate mesh-current equations for the circuit in Figure P3–10.

(b) Use these equations to find v_x and i_x.

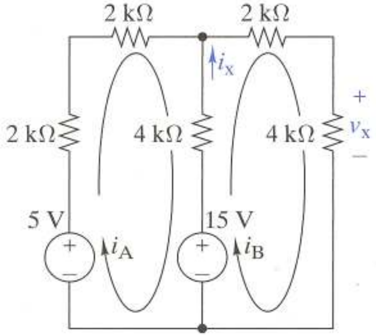

FIGURE P3–10

3–13 **(a)** Formulate mesh-current equations for the circuit in Figure P3–13.

(b) Solve for v_x and i_x when $R_1 = R_2 = 10\text{ k}\Omega$, $R_3 = 2\text{ k}\Omega$, $R_4 = 1\text{ k}\Omega$, $i_S = 2.5\text{ mA}$, $v_{S1} = 12\text{ V}$, and $v_{S2} = 0.5\text{ V}$.

(c) Find the power supplied by v_{S1}.

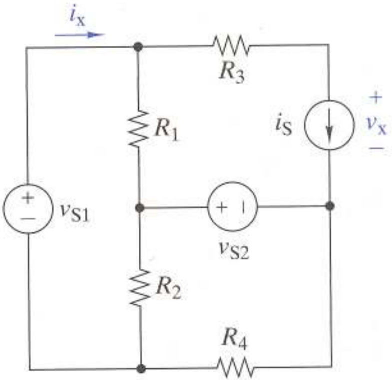

FIGURE P3–13

3-15 (a) Formulate mesh-current equations for the circuit in Figure P3–15.

(b) Use these equations to find v_x and i_x.

(c) Find the total power delivered to the resistors.

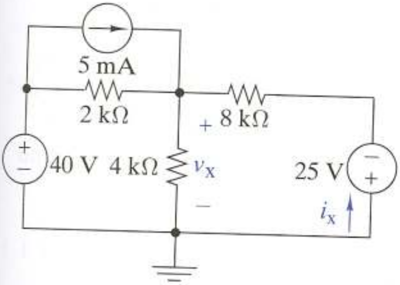

FIGURE P3–15

3-

INTEGRATING PROBLEMS

3–71 **A** Composite Circuit Analysis

The $i - v$ characteristic of network N in Figure P3–71 is $i = (5 \times 10^{-3}v - 0.5)$ A. Find the voltage at node A.

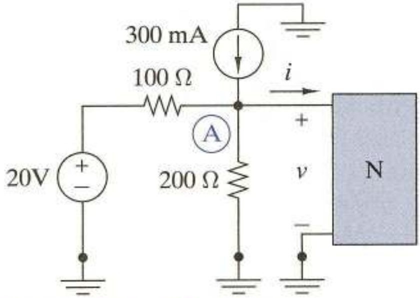

FIGURE P3–71

3–72 **A** Attenuation Analysis

In Figure P3–72 a two-port attenuator connects a 600-Ω source to a 600-Ω load. Find the power delivered to the load in terms of v_S. Remove the attenuator and find the power delivered to the load when the source is directly connected to the load. By what fraction does the attenuator reduce the power delivered to the 600-Ω load? Express the fraction in dB.

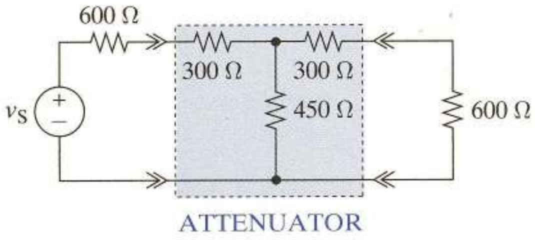

FIGURE P3–72

open-circuit voltage and Thévenin resistance requirements.

3–75 **E** Design Evaluation

A requirement exists for a circuit to deliver 0 to 5 V to a 100-Ω load from a 20-V source rated at 2.5 W. Two proposed circuits are shown in Figure P3–75. Which one would you choose and why?

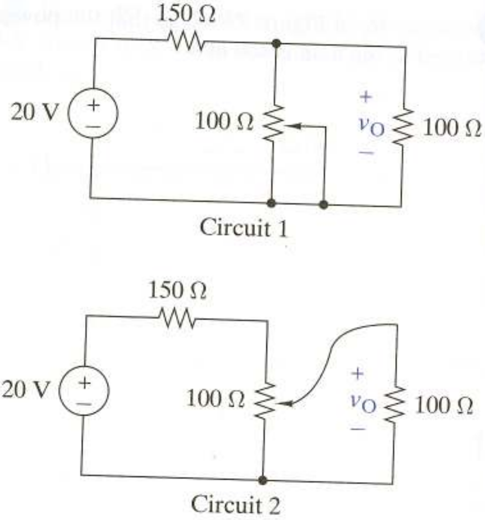

FIGURE P3–75